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As per the Module-1 Contents, the sample Q/A's are divided into as per the following ~~top~~ topics:

1. Basics (with AC quantities & 1- ϕ AC ckt.) $\rightarrow$ Q1 & Q2
 2. Phasor Notations / Phasor Diagrams $\rightarrow$ Q3, Q4, Q5
 3. Series A.C. ckt. $\rightarrow$ Q6
 4. Parallel A.C. ckt. $\rightarrow$ Q7
 5. Series Resonance $\rightarrow$ Q11, Q13
 6. Parallel Resonance $\rightarrow$ Q12, Q14
 7. Q-Factor $\rightarrow$ Q13, Q14
- Network Analysis in 1- ϕ AC ckt. $\rightarrow$ Q8, Q9, Q10

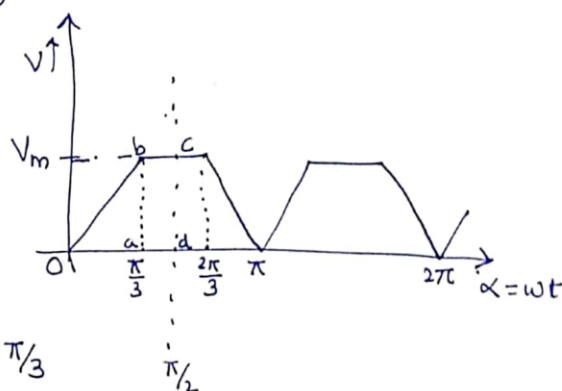
The Q/A's with these contents, are systematically presented below.

1) Basics:

Q1. Find the Avg. value of the Periodic function of the waveform below (Fig. Aside)

A. From the curve, the waveform can be written as,

$$v(t) = \begin{cases} \frac{V_m}{\pi/3}(\omega t) = \frac{V_m}{\pi/3} \alpha & \text{when } 0 \leq \alpha \leq \pi/3 \\ V_m & \text{when } \pi/3 \leq \alpha \leq \pi/2 \end{cases}$$



The given periodic function has time period 'T' = $\frac{\pi}{\omega}$ sec. It repeats at intervals of ' π ' = π radians.

Avg. value of the periodic $f^n = \frac{1}{T} \int_0^T f(t) dt$

(ex) $F_{av.} = \frac{1}{T} \int_0^T f(\omega t) d(\omega t)$ where T is expressed in radians.

$$\begin{aligned}
 \therefore V_{av.} &= \frac{1}{\pi} \int_0^{\pi} v(\omega t) d(\omega t) \\
 &= \frac{1}{\pi} \left[\underbrace{\int_0^{\pi/3} \frac{V_m}{(\pi/3)} \cdot \alpha d\alpha}_{\text{wave symmetry}} + \int_{\pi/3}^{\pi/2} V_m d\alpha \right. \\
 &\quad \left. + \int_{\pi/2}^{2\pi/3} V_m d\alpha + \int_{2\pi/3}^{\pi} \frac{V_m}{(\pi/3)} (\pi - \alpha) d\alpha \right] \\
 &= \frac{2}{\pi} \left[\int_0^{\pi/3} \frac{V_m}{(\pi/3)} \cdot \alpha d\alpha + \int_{\pi/3}^{\pi/2} V_m d\alpha \right] \\
 &= \frac{2}{3\pi} \left[\frac{V_m}{(\pi/3)} \left[\frac{\alpha^2}{2} \right]_0^{\pi/3} + V_m [\alpha]_{\pi/3}^{\pi/2} \right] \\
 &= \frac{2V_m}{\pi} \left[\frac{\pi}{6} + \frac{\pi}{6} \right] = \frac{2}{3} V_m
 \end{aligned}$$

→ Result can be validated by finding area under curve from '0' to ' $\pi/2$ ' & dividing by ' $\pi/2$ '.

$$\text{Area} |_{v(t) \text{ from '0' to '}\pi/2} = \text{Area} |_{oab} + \text{Area} |_{abcd} = \frac{1}{2} V_m \left(\frac{\pi}{3} \right) + V_m \left(\frac{\pi}{6} \right) = \frac{\pi}{3} V_m$$

$$\therefore V_{av} = \frac{\text{area}}{(\pi/2)} = \frac{(\pi/3) V_m}{(\pi/2)} = \frac{2}{3} V_m$$

(2)

Q2. A voltage $v(t) = 170 \sin(314t + 10^\circ)$ is applied to a circuit. It causes a steady-state current, $i(t) = 14.14 \sin(314t - 20^\circ)$ to flow in ckt. Find the Power Factor and Avg. Power delivered to the circuit.

A. Comparison of expressions $v(t)$ & $i(t)$ reveals that, the phase-difference b/n v & $i = 30^\circ$

$$\therefore \text{Power Factor} = \cos \theta = \cos 30^\circ = \underline{0.866}$$

$$\text{Also, } V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}} = \frac{170}{\sqrt{2}} = 120\text{V}$$

$$I_{\text{rms}} = \frac{I_{\text{peak}}}{\sqrt{2}} = \frac{14.14}{\sqrt{2}} = 10\text{A}$$

$$\therefore \text{Avg. Power / Active Power} = P_{\text{av}} = VI \cos \theta = 120 \times 10 \times 0.866 = \underline{1040\text{W}}$$

2) Phasor Notations/Diags:

Q3. The two sinusoidal currents are given as,

$$i_1 = 10\sqrt{2} \sin \omega t \quad i_2 = 20\sqrt{2} \sin(\omega t + 60^\circ)$$

Find expression of sum of these currents using Phasor Addition.

A. i_1 & i_2 have RMS values of 10A & 20A (Ref. Phasor)

$$\text{i.e. } i_{1\text{rms}} = 10\text{A} \Rightarrow i_{1\text{peak}} = |I_1| = 10\sqrt{2} ; \text{Angle} = 0^\circ$$

$$i_{2\text{rms}} = 20\text{A} \Rightarrow i_{2\text{peak}} = |I_2| = 20\sqrt{2} ; \text{Angle} = 60^\circ$$

Therefore, the phasor quantity of currents are,
(in terms of peak)

$$\bar{I}_1 = 10\sqrt{2} (\cos 0^\circ + j \sin 0^\circ) \\ = \underline{10\sqrt{2}}$$

$$\bar{I}_2 = 20\sqrt{2} (\cos 60^\circ + j \sin 60^\circ) \\ = 20\sqrt{2} \left(\frac{1}{2} + j \frac{\sqrt{3}}{2} \right) = 10\sqrt{2} + j 10\sqrt{6}$$

Thus, $\bar{I}_{net} = \bar{I}_1 + \bar{I}_2 = 10\sqrt{2} + (10\sqrt{2} + j 10\sqrt{6})$
 $= 20\sqrt{2} + j 10\sqrt{6}$

$$\Rightarrow |\bar{I}_{net}| = \sqrt{(20\sqrt{2})^2 + (10\sqrt{6})^2} = \sqrt{800 + 600} \\ \text{(magnitude)} \quad \quad \quad = \underline{37.4} \\ \text{Peak.}$$

$$\text{Angle of } \bar{I}_{net} \text{ i.e. } \theta_{net} = \tan^{-1} \left(\frac{10\sqrt{6}}{20\sqrt{2}} \right) = \tan^{-1} \frac{\sqrt{3}}{2} \\ = \underline{41^\circ}$$

$$\therefore \underline{\bar{I}_{net} = 37.4 e^{j41^\circ}}$$

$$\Rightarrow \boxed{i_{net} = 37.4 \sin(\omega t + 41^\circ)}$$

Note: In interest of simplicity, it is customary to replace

$$e^{j\theta} \text{ by } \angle \theta ; \text{ i.e. } \bar{I}_{net} = \frac{37.4}{\sqrt{2}} e^{j41^\circ} = \frac{37.4}{\sqrt{2}} \angle 41^\circ \\ \text{(RMS)}$$

Q4. A sinusoidal having effective/RMS value of $10 \angle 0^\circ$ A is added to sinusoidal current $20 \angle 60^\circ$ A. Find resultant current. (3)

A.
$$\begin{aligned}\bar{I}_{\text{net}} &= \bar{I}_1 + \bar{I}_2 = 10 \angle 0^\circ + 20 \angle 60^\circ \\ &= 10 + (10 + j10\sqrt{3}) = 20 + j10\sqrt{3} \\ &= 26.4 \angle 41^\circ \text{ A}\end{aligned}$$

So, for complete time expression,

$$\begin{aligned}i_{\text{net}} &= \sqrt{2} \cdot 26.4 \sin(\omega t + 41^\circ) \\ &= 37.4 \sin(\omega t + 41^\circ)\end{aligned}$$

→ This is exactly equal to the i_{net} obtained in answer to Q3.

Note: In view of this, it is customary to use a phasor diagram in which the phasors are expressed as effective/RMS values, rather than using their maximum values.

Thus, phasor $10 \angle 0^\circ \Rightarrow$ ^{Peak} magnitude of $10\sqrt{2}$, with RMS value 10A.
 $\neq$ " $20 \angle 60^\circ \Rightarrow$ " " $20\sqrt{2}$, " " 20A.
RMS

(This convention shall be followed in this course)

Q5. Evaluate the expression below using Phasor method:

$$e(t) = 100\sqrt{2} \cos(314t - 30^\circ) \pm 200\sqrt{2} \sin(314t - 60^\circ)$$

For both '+' & '-' cases.

A.
$$e(t) = 100\sqrt{2} \cos(314t - 30^\circ) \pm 200\sqrt{2} \sin(314t - 60^\circ)$$

Case $\begin{cases} \text{(I)} & e(t) = 100\sqrt{2} \cos(314t - 30^\circ) + 200\sqrt{2} \sin(314t - 60^\circ) \\ \text{(II)} & e(t) = 100\sqrt{2} \cos(314t - 30^\circ) - 200\sqrt{2} \sin(314t - 60^\circ) \end{cases}$

Case-I:

$$\begin{aligned} e(t) &= 100\sqrt{2} \cos(314t - 30^\circ) + 200\sqrt{2} \sin(314t - 60^\circ) \\ &= 100\sqrt{2} \cos(314t - 30^\circ) + 200\sqrt{2} \cos(90^\circ - (314t - 60^\circ)) \\ &= 200\sqrt{2} (-\cos(314t + 30^\circ)) \end{aligned}$$

$$\text{wt } e(t) \equiv E \angle \theta = \bar{E}$$

$$\begin{aligned} \Rightarrow \bar{E} &= 100 \angle -30^\circ - 200 \angle +30^\circ \\ &= 100 \left(\frac{\sqrt{3}}{2} - j \cdot \frac{1}{2} \right) - 200 \left(\frac{\sqrt{3}}{2} + j \cdot \frac{1}{2} \right) \\ &= -100 \cdot \frac{\sqrt{3}}{2} - 300 \cdot \frac{1}{2} \cdot j \\ &= -86.6 - 150j = 173.2 \angle -120^\circ \\ \Rightarrow e_1(t) &= 173.2\sqrt{2} \cos(\omega t - 120^\circ) \end{aligned}$$

Case-II:

$$\begin{aligned} e(t) &= 100\sqrt{2} \cos(314t - 30^\circ) + 200\sqrt{2} \sin(314t - 60^\circ) \\ &= 100\sqrt{2} \cos(314t - 30^\circ) + 200\sqrt{2} \cos(314t + 30^\circ) \end{aligned}$$

$$\begin{aligned} \therefore \bar{E} &= 100 \angle -30^\circ + 200 \angle +30^\circ \\ &= 100 \left(\frac{\sqrt{3}}{2} - j \cdot \frac{1}{2} \right) + 200 \left(\frac{\sqrt{3}}{2} + j \cdot \frac{1}{2} \right) \\ &= 300 \frac{\sqrt{3}}{2} + 100 \cdot \frac{1}{2} \cdot j \\ &= 259.8 + 50j = 264.5 \angle 10.53^\circ \\ \Rightarrow e_2(t) &= 264.5\sqrt{2} \cos(\omega t + 10.53^\circ) \end{aligned}$$

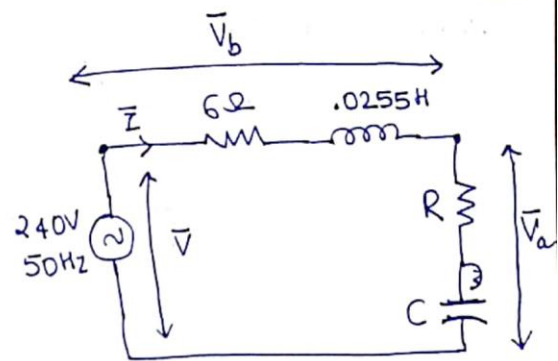
3) Series AC Ckt:

Q6. In the circuit shown aside,

Given that $\bar{V}_b \perp \bar{V}_a$ are in quadrature, & $|V_b| = 3|V_a|$,

Find 'R' & 'C'. Also depict the

Phasor relation between V, V_b, V_a & $\bar{I}$, with Phasor Diagram.



A. $240V, 50Hz \Rightarrow |V|_{rms} = 240$
Supply.

It is said that $\bar{V}_b \perp \bar{V}_a$ are in Quadrature.

$$L = 0.0255 \text{ H}$$

$$\Rightarrow X_L = \omega L$$

$$= 2 \times 3.14 \times 50 \times L$$

$$= 8 \underline{\underline{\Omega}}$$

$$\Rightarrow |\bar{V}_a|_{rms}^2 + |\bar{V}_b|_{rms}^2 = |\bar{V}|_{rms}^2 \quad (\text{from Fig.})$$

$$\& |V_b| = 3|V_a|$$

$$\Rightarrow |V_a|_{rms}^2 + 9|V_a|_{rms}^2 = |\bar{V}|_{rms}^2$$

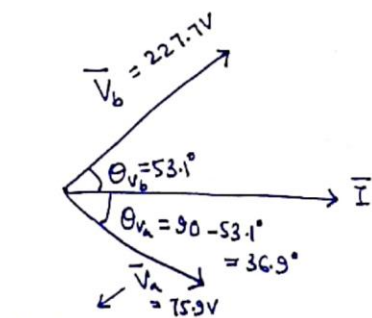
$$\Rightarrow 10|V_a|_{rms}^2 = 240^2 \Rightarrow$$

$$|V_a|_{rms} = 75.9 \text{ V.}$$

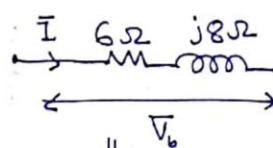
$$|V_b|_{rms} = 3|V_a|_{rms}$$

$$= 227.7 \text{ V}$$

$\therefore$ Phasor Diagram taking $\bar{I}$ as reference phasor:



$\bar{V}_a$ lags $\bar{I}$ because of Capacitive element



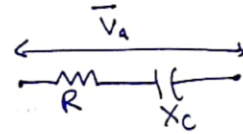
$\bar{V}_b$ leads $\bar{I}$ ($\because$ Inductive element)

with angle, $\theta_v = \tan^{-1}\left(\frac{8}{6}\right)$

$$= 53.1^\circ$$

∴ From the phasor above,

$$\theta_{V_a} = 36.9^\circ = \tan^{-1} \left(\frac{X_c}{R} \right)$$



$$\Rightarrow \frac{X_c}{R} = \tan 36.9^\circ = 0.75 \Rightarrow \underline{X_c = 0.75 R} \quad \text{--- (1)}$$

$$\text{Also, } |\bar{V}_b| = 227.7 \text{ V} \quad |\bar{I}| = \frac{|\bar{V}_b|}{|Z|} = \frac{227.7}{\sqrt{6^2 + 8^2}} = \underline{22.8 \text{ A}}$$

∴

$$|\bar{I}| \times |\bar{V}_a| \times |Z_c| = |\bar{V}_a| \quad (\because \text{Series circuit})$$

$$\Rightarrow 22.8 \cdot \sqrt{R^2 + X_c^2} = 75.9$$

$$\Rightarrow R^2 + X_c^2 = \left(\frac{75.9}{22.8} \right)^2 = 11.09 \quad \text{--- (2)}$$

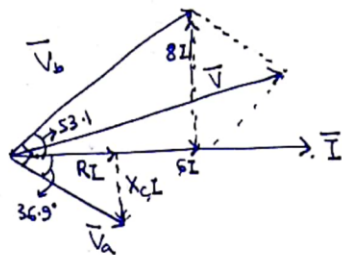
Solving (1) & (2),

$$\boxed{R = 2.664 \Omega}$$

$$X_c = 1.998 \Omega$$

$$\& X_c = \frac{1}{\omega C} = \frac{1}{2\pi \times 3.14 \times 50 \times C} \Rightarrow \boxed{C = 1.594 \text{ mF}}$$

So,



(overall phasor Diagram)

4) Parallel AC Ckt:

(5)

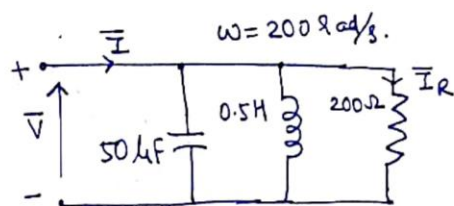
Q7. In the parallel circuit Fig. aside,

(a) Find $\bar{I}$, if $\bar{I}_R = 0.02 \angle 30^\circ \text{ A}$

(b) Find $\bar{I}_R$, if $\bar{I} = 2 \angle -40^\circ \text{ A}$;

also find the applied voltage ($\bar{V}$).

— Also, Identify, with associated phasor diagram, phasors $\bar{I}_R, \bar{I}_L, \bar{I}_C$ & $\bar{V}$ in (a).



A.

(a) Given $\bar{I}_R = .02 \angle 30^\circ \text{ A}$.

$$\Rightarrow \text{Applied voltage, } \bar{V} = 200 \Omega \times .02 \angle 30^\circ \\ = 4 \angle 30^\circ \text{ V.}$$

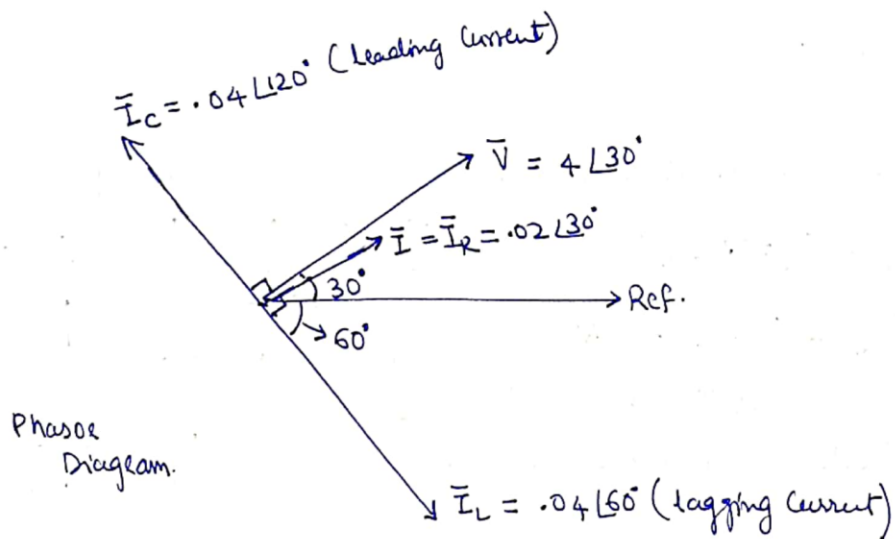
$$\Rightarrow \bar{I}_L = \frac{4 \angle 30^\circ}{j200 \times 0.5} = .04 \angle -60^\circ \text{ A}$$

$$\Rightarrow \bar{I}_C = \frac{4 \angle 30^\circ}{\left(\frac{1}{j200 \times 50 \times 10^{-6}} \right)} = .04 \angle 120^\circ \text{ A.}$$

$$\begin{aligned} \therefore \bar{I} &= \bar{I}_R + \bar{I}_L + \bar{I}_C \\ &= .02 \angle 30^\circ + .04 \angle -60^\circ + .04 \angle 120^\circ \\ &= (.0173 + j0.01) + (.02 - j0.0346) + (-0.02 + j0.0346) \\ &= .0173 + j0.01 = \underline{\underline{0.02 \angle 30^\circ \text{ A}}} \end{aligned}$$

Note that $\bar{I}_L$ & $\bar{I}_C$ phasors, cancel-out in a way so that,
 $\bar{I} = \bar{I}_R$ eventually,

The phasor diagram of the same is as follows,



(b) Since, parallel ckt,

$$\frac{1}{Z_{net}} = \frac{1}{200} + \frac{1}{j200 \times 0.5} + \frac{1}{\frac{1}{j200 \times 50 \times 10^{-6}}}$$

$$= .005 - j0.01 + j0.01 = 0.005$$

$$\Rightarrow Z_{net} = 200 \Omega$$

$$\therefore \bar{V} = 200 \times 2 \angle -40^\circ \quad (\text{Given } \bar{I} = 2 \angle -40^\circ)$$

$$= 400 \angle -40^\circ$$

$$\Rightarrow \bar{I}_R = \frac{\bar{V}}{R} = \frac{400 \angle -40^\circ}{200} = \underline{\underline{2 \angle -40^\circ}}$$

$$\& \text{ Applied Voltage } \underline{\underline{\bar{V} = 400 \angle -40^\circ}}$$

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Network Analysis in 1- ϕ AC Ckts:

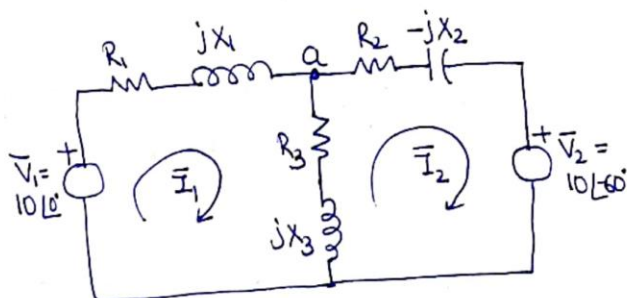
Q8. (Mesh Method of Analysis) The two-mesh network with sinusoidal voltage source in each mesh is as shown in Fig. below. The

Impedances are given as,

$$\bar{Z}_1 = R_1 + jX_1 = 1 + j = \sqrt{2} \angle 45^\circ$$

$$\bar{Z}_2 = R_2 - jX_2 = 1 - j = \sqrt{2} \angle -45^\circ$$

$$\bar{Z}_3 = R_3 + jX_3 = 1 + j2 = \sqrt{5} \angle 63.5^\circ$$



$$\bar{V}_1 = 10 \angle 0^\circ \quad \& \quad \bar{V}_2 = 10 \angle -60^\circ$$

Find the phasor expressions for currents through branches $\bar{Z}_1$, $\bar{Z}_2$ & $\bar{Z}_3$.

A. First step: Write the Mesh Eq's in terms of Phasor Quantities.

$$\text{So, } \bar{V}_1 - \bar{I}_1 \bar{Z}_1 - \bar{Z}_3 (\bar{I}_1 - \bar{I}_2) = 0 \rightarrow \text{in Mesh 1}$$

$$\Rightarrow \underbrace{\bar{I}_1 (\bar{Z}_1 + \bar{Z}_3)}_{\bar{Z}_{11}} - \underbrace{\bar{I}_2 \bar{Z}_3}_{\bar{Z}_{12}} = \bar{V}_1 \quad \text{--- (1)}$$

$$\text{Similarly, } -\bar{Z}_3 (\bar{I}_2 - \bar{I}_1) - \bar{Z}_2 \bar{I}_2 - \bar{V}_2 = 0 \rightarrow \text{in Mesh 2}$$

$$\Rightarrow -\underbrace{\bar{I}_1 \bar{Z}_3}_{\bar{Z}_{21}} + \underbrace{\bar{I}_2 (\bar{Z}_2 + \bar{Z}_3)}_{\bar{Z}_{22}} = -\bar{V}_2 \quad \text{--- (2)}$$

From (1) & (2),
upon solving,

$$\bar{I}_1 = \frac{\bar{V}_1 \bar{Z}_{22} - \bar{V}_2 \bar{Z}_{12}}{\bar{Z}_{11} \bar{Z}_{22} - \bar{Z}_{12}^2} = \frac{\bar{V}_1 (\bar{Z}_2 + \bar{Z}_3) - \bar{V}_2 (\bar{Z}_3)}{(\bar{Z}_1 + \bar{Z}_3)(\bar{Z}_2 + \bar{Z}_3) - \bar{Z}_3^2}$$

$$= \frac{10 \angle 0^\circ (1-j+1+2j) - 10 \angle -60^\circ (1+2j)}{(1+j+1+2j)(1-j+1+2j) - (1+2j)^2}$$

Substituting $10 \angle 0^\circ = 10$ & $10 \angle -60^\circ = (5 - 5\sqrt{3}j)$ & Solving,

$$\text{we get, } \bar{I}_1 = \frac{20 + 10j - 22.32 - 1.34j}{4 + 4j} = \frac{8.97 \angle 105^\circ}{5.66 \angle 45^\circ}$$

$$= 1.58 \angle 60^\circ$$

$$\therefore \boxed{\bar{I}_1 = 1.58 \angle 60^\circ \text{ A} = 0.79 + j1.37 \text{ A}} \rightarrow \text{Current through } Z_1$$

$$\text{Similarly, } \bar{I}_2 = \frac{-\bar{V}_2 Z_{11} + Z_{21} \bar{V}_1}{Z_{11} Z_{22} - Z_{12}^2} = \frac{-\bar{V}_2 (Z_1 + Z_3) + \bar{V}_1 Z_3}{(Z_1 + Z_3)(Z_2 + Z_3) - Z_3^2}$$

Substituting Z_1, Z_2, Z_3 values & $\bar{V}_1 = 10; \bar{V}_2 = 5 - 5\sqrt{3}j$,

$$\text{we get, } \bar{I}_2 = \frac{34.1 \angle 139.3^\circ}{5.66 \angle 45^\circ} = 6.03 \angle 94.3^\circ$$

$$\therefore \boxed{\bar{I}_2 = 6.03 \angle 94.3^\circ \text{ A} = -0.46 + j6.01 \text{ A}} \rightarrow \text{Current phasor through } Z_2$$

Current phasor through Z_3 is,

$$\bar{I}_1 - \bar{I}_2$$

$$= (0.79 + j1.37) - (-0.46 + j6.01)$$

$$= 1.25 - 4.64j = \underline{\underline{4.82 \angle -75^\circ \text{ A}}}$$

Q9. (Nodal Method Analysis) . Find the voltage at Node 'a' in the (7)

Circuit of 'Q8' with all the same given data.

Also find current in branch Z_3 with this Nodal Analysis.

A. Assuming currents flow away from Node 'a' & applying the Nodal (KCL) Eqⁿ at Node 'a',

$$\frac{\bar{V}_a - \bar{V}_1}{\bar{Z}_1} + \frac{\bar{V}_a}{\bar{Z}_3} + \frac{\bar{V}_a - \bar{V}_2}{\bar{Z}_2} = 0$$

$$\Rightarrow \bar{V}_a \left(\frac{1}{\bar{Z}_1} + \frac{1}{\bar{Z}_3} + \frac{1}{\bar{Z}_2} \right) = \frac{\bar{V}_1}{\bar{Z}_1} + \frac{\bar{V}_2}{\bar{Z}_2}$$

$$\begin{aligned} \frac{1}{\bar{Z}_1} + \frac{1}{\bar{Z}_2} + \frac{1}{\bar{Z}_3} &= \frac{1}{1+j} + \frac{1}{1-j} + \frac{1}{1+2j} \\ &= \frac{1}{2} - j\frac{1}{2} + \frac{1}{2} + j\frac{1}{2} + \frac{1}{5} - j\frac{2}{5} = 1.2 - 0.4j \\ &= 1.263 \angle -18.4^\circ \end{aligned}$$

$$\begin{aligned} \frac{\bar{V}_1}{\bar{Z}_1} + \frac{\bar{V}_2}{\bar{Z}_2} &= \frac{10 \angle 0^\circ}{\sqrt{2} \angle 45^\circ} + \frac{10 \angle -60^\circ}{\sqrt{2} \angle -45^\circ} = 5 - j5 + 6.81 - j1.83 \\ &= 11.81 - j6.83 \\ &= 13.65 \angle -30^\circ \end{aligned}$$

Thus, $\bar{V}_a \cdot (1.263 \angle -18.4^\circ) = 13.65 \angle -30^\circ$

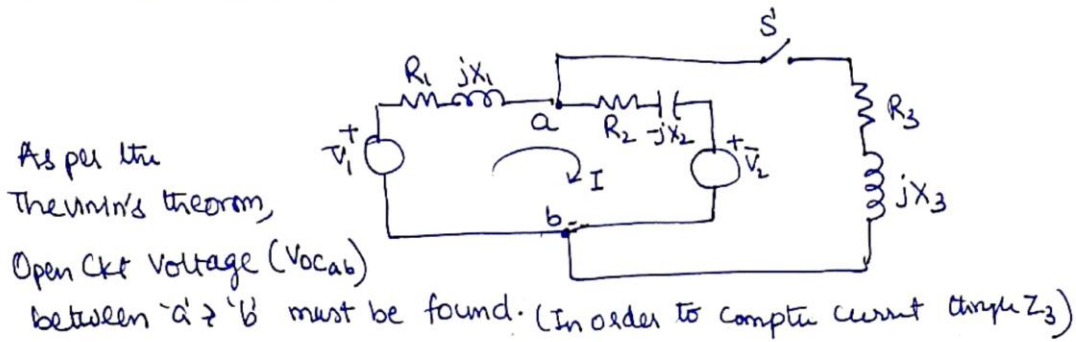
or, $\bar{V}_a = \frac{13.65 \angle -30^\circ}{1.263 \angle -18.4^\circ} = 10.8 \angle -11.6^\circ \text{ V}$

Thus, $I_{\text{thru } Z_3} = \frac{\bar{V}_a}{\bar{Z}_3} = \frac{10.8 \angle -11.6^\circ}{\sqrt{5} \angle 63.5^\circ} = 4.82 \angle -75^\circ$

This validates the result in 'Q8' as well.

Q10. (Solve by Thevenin's Theorem) Find the current through the branch ' Z_3 ' in the Fig. with 'Q8', with all the same given data, by using thevenin's theorem.

A. The circuit can be re-drawn as follows.



So, after opening S' , & applying KVL,

$$\Rightarrow \bar{V}_1 - Z_1 \bar{I} - Z_2 \bar{I} - \bar{V}_2 = 0$$

$$\Rightarrow \bar{I} = \frac{(\bar{V}_1 - \bar{V}_2)}{(Z_1 + Z_2)} = \frac{10 \angle 0^\circ - 10 \angle -60^\circ}{1 + j + 1 - j} = \frac{10 - 5 + j8.66}{2} = \underline{\underline{5 \angle 60^\circ}}$$

$$\therefore V_{ocab} = \bar{V}_1 - \bar{I} Z_1$$

$$= 10 \angle 0^\circ - 5 \angle 60^\circ \cdot (\sqrt{2} \angle 45^\circ) = 10 + 1.83 - j6.82 = \underline{\underline{13.65 \angle -30^\circ}}$$

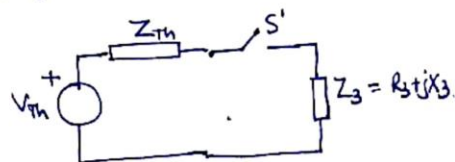
& Thevenin's Impedance: (Z_{Th})

after short-circuiting voltage sources, the Equivalent Impedance between 'a' & 'b' can be expressed as,

$$Z_{Th} = \frac{Z_1 Z_2}{Z_1 + Z_2} = \frac{\sqrt{2} \angle 45^\circ \cdot \sqrt{2} \angle -45^\circ}{1 + j + 1 - j} = \underline{\underline{1 \angle 0^\circ \Omega}}$$

Note: This voltage has larger magnitude than any of source voltage.

$$\therefore \bar{I}_{\text{through } Z_3} = \frac{\bar{V}_{Th}}{Z_{Th} + Z_3} = \frac{13.65 \angle -30^\circ}{1 + 1 + 2j} = \underline{\underline{4.82 \angle -75^\circ \text{ A}}}$$



This validates result in 'Q8' & 'Q9'!

5) Series Resonance:

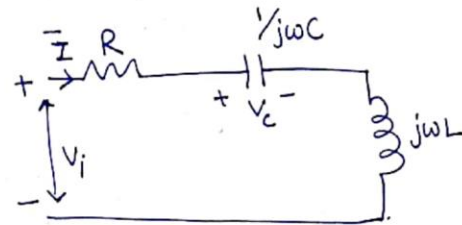
(8)

Q11. A Capacitor of 12 nF is connected in series with an inductor of 4 mH and 5Ω resistance as shown below/side:

(a) Calculate Resonant Frequency, ω_0 .

(b) at ω_0 , If $|V_C| = 1.5 \text{ V}$, what is $|\bar{V}_i|$?

(c) Along with using Phasor diagram, show how Capacitor voltage can be larger than the Input voltage ($\bar{V}_i$).



A. (a) Resonant Frequency,

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4 \times 10^{-3} \times 12 \times 10^{-9}}} = 0.144 \times 10^6 = \underline{\underline{144 \text{ k rad/s.}}}$$

(b) At ω_0 , $V_C = 1.5 \text{ V}$

$$\text{So, } |\bar{I}| = |\bar{V}_C| / \omega_0 C = 1.5 \times 144 \times 10^3 \times 12 \times 10^{-9} = 2.592 \text{ mA}$$

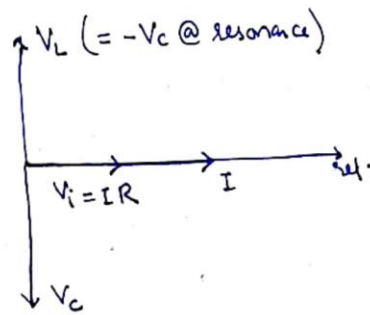
$$\text{So, } V_i = R \bar{I} \quad (\because \text{ @ resonance, Circuit Impedance = circuit Resistance})$$

$$= 5 \times 2.592 = \underline{\underline{12.96 \text{ mV}}}$$

(c) At ω_0 , $\bar{I}$ will be in phase with $\bar{V}_i$

& $\bar{V}_C$ lags $\bar{I}$ by 90° ; $\bar{V}_L$ leads $\bar{I}$ by 90°

So,



$$\text{Capacitive reactance, } X_C(\Omega) = \frac{1}{\omega_0 C} = \frac{1}{144 \times 10^3 \times 12 \times 10^{-9}}$$

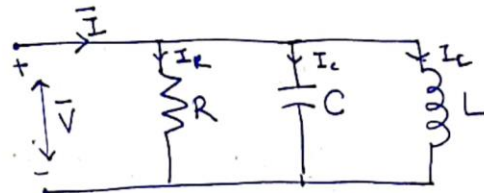
$$= 579 \Omega \quad \text{@ resonance Freq. } \omega_0$$

$$\therefore X_C \gg 5 \Omega ; \text{ Therefore } V_C \gg V_R$$

6) Parallel Resonance:

Q12. In the circuit as shown in Fig. A side, $R = \frac{1}{G} = 0.2 \text{ M}\Omega$, $L = 2 \text{ mH}$,
it draws minimum current at freq. = 5 k rad/s , with min. current = 2 mA .

- (a) At what value of 'C',
the voltage $\bar{V}$ is maximum?
& what is its value?



- (b) with the value of 'C' in (a),
what should be the value of $\bar{I}$ for inductor current of 1 A

A. For $\bar{V}$ to be maximum, the ckt must be under resonance,

$$\Rightarrow \omega_0^2 = \frac{1}{LC} \quad \text{or} \quad C = \frac{1}{\omega_0^2 L}$$

$$\therefore C = \frac{1}{(5 \times 10^3)^2 \cdot 2 \times 10^{-3}} = 20 \mu\text{F}$$

(9)

Under resonance, $\bar{I}$ flows through 'R' only; So,

$$\begin{aligned} |\bar{V}| &= |\bar{I}|R = |\bar{I}| \times 0.2 \times 10^6 \\ &= 2 \times 10^{-3} \times 0.2 \times 10^6 \\ &= \underline{\underline{400 \text{ V}}} \end{aligned}$$

(b) Given $L = 2 \text{ mH}$

$$\Rightarrow \omega_0 L = 5 \times 10^3 \times 2 \times 10^{-3} = 10 \Omega.$$

($= X_L$)

$$\begin{aligned} \therefore \bar{I}_L &= \frac{\bar{V}}{X_L} = \frac{\bar{V}}{\omega_0 L} = \frac{\bar{I} R}{\omega_0 L} \\ &= \frac{0.2 \times 10^6 \bar{I}}{10} \end{aligned}$$

$$\begin{aligned} \therefore \text{If } \bar{I}_L &= 1 \text{ A}, \Rightarrow \bar{I} = \frac{10}{0.2 \times 10^6} \times 1 \text{ A} \\ &= 50 \times 10^{-6} \text{ A} = \underline{\underline{0.05 \text{ mA}}} \end{aligned}$$

7) Q-Factor (along with Series/Parallel Resonance):

Q13. A voltage with frequency 1 MHz is connected across an inductor (L) (Const. Sinusoidal source) having some internal resistance (r), in series with a variable capacitor. The circuit draws maximum current when the capacitance is 500 pF . The current reduces by a factor of $(\frac{1}{\sqrt{2}})$ when capacitor is adjusted to 450 pF . determine r , X_L & Q_0 (Q-Factor).

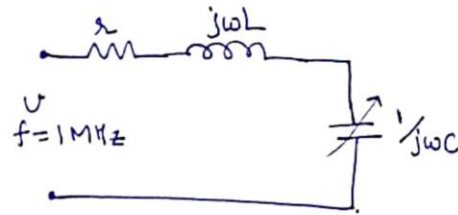
A. The circuit is drawn as shown side:

When

$$C = 500 \text{ pF},$$

Current drawn is max. @ 1 MHz.

i.e. resonant freq. = 1 MHz.



$$\Rightarrow \omega_0^2 = \frac{1}{LC} \Rightarrow L = \frac{1}{\omega_0^2 C} = \frac{1}{(2\pi \times 10^6)^2 \cdot 500 \times 10^{-12}}$$

$$= \underline{\underline{.0633 \text{ mH}}}$$

When

$$C = 450 \text{ pF},$$

the resonant Freq. of ckt becomes, $\omega_0' = \frac{1}{\sqrt{LC}}$

$$= \frac{1}{\sqrt{.0633 \times 10^{-3} \times 450 \times 10^{-12}}}$$

$$= 5.925 \times 10^6 \text{ rad/s.}$$

$$\Rightarrow f_0' = \frac{\omega_0'}{2\pi} = \underline{\underline{.943 \text{ MHz}}}$$

$\therefore$ With applied frequency of 1 MHz, & $C = 450 \text{ pF}$

Current reduces by $\frac{1}{\sqrt{2}}$ $\therefore f_0'$ becomes the half power frequency.

$$\frac{|I|}{I_2} = \frac{|V|}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}; |I| = \frac{|V|}{R}.$$

$$@ \omega = 2\pi \times 1 \text{ MHz} \text{ and } C = 450 \text{ pF}.$$

$$\Rightarrow R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 = 2 \frac{|V|^2}{|I|^2} = 2 R^2$$

$$\Rightarrow \omega L - \frac{1}{\omega C} = R @ \omega = 2\pi \times 10^6 \text{ rad/s}$$

$$\& C = 450 \times 10^{-12} \text{ F.}$$

$$\text{while } L = .0633 \text{ mH}$$

(10)

$$\Rightarrow 2\pi \times 10^6 \times 0.0633 \times 10^{-3} - \frac{1}{2\pi \times 10^6 \times 450 \times 10^{-12}} = 2$$

$$\Rightarrow \underline{2 \approx 44 \Omega}$$

Quality Factor, $Q_0 = \frac{\omega_0 L}{R}$

$$= \frac{2\pi \times 10^6 \times 0.0633 \times 10^{-3}}{44} \approx \underline{9}$$

Q14. An RLC parallel circuit has $R = 1 \text{ M}\Omega$, $L = 1 \text{ H}$, $C = 1 \mu\text{F}$.

It is excited by a current $I = 10 \angle 0^\circ \text{ A}$. Find ω_0 and Q_0 .
↓
Resonant Freq.

If freq. of current I is ω_0 , find voltage across the circuit.

At what frequencies, the voltage reduces by $(1/\sqrt{2})$?

A.

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 10^{-6}}} = 10^3 \text{ rad/s.}$$

At resonance, current I flows only through 'R'. Therefore,

$$\underline{V} = \underline{I} R = 10 \times 10^{-6} \times 1 \times 10^6 = 10 \angle 0^\circ \text{ V}$$

↑
voltage across ckt when I has freq. ω_0 .

→ The voltage gets reduced by $(1/\sqrt{2})$ at the
Half-power frequencies

Therefore, Firstly, Quality factor $Q_0 = \omega_0 RC$ (In parallel RLC circuit)

$$= 10^3 \times 1 \times 10^6 \times 1 \times 10^{-6}$$

$$= \underline{\underline{10^3}}.$$

Band-width,

$$\omega_b = \frac{\omega_0}{Q_0} = \frac{10^3}{10^3} = 1 \text{ rad/s.}$$



$\therefore$ Frequencies at which voltage reduces by $1/\sqrt{2}$ is,

$$\omega_0 \pm \frac{\omega_b}{2}$$

$$\therefore \omega_1 = \omega_0 - \frac{\omega_b}{2} = 10^3 - \frac{1}{2} = \underline{\underline{999.5 \text{ rad/s.}}}$$

$$\omega_2 = \omega_0 + \frac{\omega_b}{2} = 10^3 + \frac{1}{2} = \underline{\underline{1000.5 \text{ rad/s.}}}$$

Best Wishes
②
 Vishal